

Emergency Alerting Systems Overview

Purpose

The purpose of this research document is to discuss the characteristics, capabilities and limitations of the existing Emergency Alerting Systems in the United States. This information will be used to determine if a similar system could be implemented for the Rail Crossing Violation Warning (RCVW) System.

Background

The first Emergency Alerting system (AMBER Alert) was created by the state of Texas following the 1996 abduction and murder of 9-year-old Amber Hagerman. Broadcasters teamed with law enforcement agencies to create an early warning system to aid in locating missing children.

In May 2005, the Department of Justice joined the wireless industry and other government officials to announce that wireless users can opt to receive geographically specified AMBER Alert messages on their wireless devices through an AMBER Alert wireless messaging system.

On December 31, 2012, the AMBER Alert program is retired and replaced with the Wireless Emergency Alerts (WEA) system. This system allows government officials to issue emergency alerts such as

- National Alerts – Issued by the President of the United States or the Administrator of FEMA
- Imminent Threats – Alerts involving imminent threats to safety or life
- AMBER Alerts – Missing children alerts
- Public Safety Messages – Alerts conveying recommendations for saving lives and property

The Wireless Emergency Alerts system is an essential part of America's emergency preparedness. Since its launch in 2012, the WEA system has been used more than 70,000 times to warn the public about dangerous weather, missing children, and other critical situations – all through alerts on compatible cell phones and other mobile devices. WEA enables government officials to target emergency alerts within specific geographic areas. Wireless companies volunteer to participate in WEA, which is the result of a unique public/private partnership between the Federal Emergency Management Agency, the FCC, and the United States wireless industry in order to enhance public safety. However, even a participating wireless company may only offer WEA in specific areas.

On October 31, 2012, Google integrated AMBER alerts into their Public Alerts service and search engine and on March 3, 2015, it integrated AMBER alerts into the Waze traffic app.

General Requirements to the issuing of an Alert

In the United States, AMBER alerts are distributed via commercial and public radio stations, Internet radio, satellite radio, television stations, mobile phones and cable TV by the Emergency Alert System. Once law enforcement has determined that a child has been abducted and the abduction meets AMBER Alert criteria, law enforcement notifies broadcasters and state transportation officials.

Authorized national, state, or local government authorities send alerts regarding public safety emergencies using WEA. Authorized public safety officials send WEA alerts through FEMA's Integrated Public Alert and Warning System (IPAWS) to participating wireless carriers, which then push the alerts to compatible mobile devices in the affected area. A WEA alert appears on the screen of the recipient's

handset as a text-like message. The alert is accompanied by a unique attention signal and vibration, which is particularly helpful to people with hearing or vision-related disabilities.

In order to issue an AMBER alert, each of the following has to occur:

- There is reasonable belief by law enforcement that an abduction has occurred.
- The law enforcement agency believes that the child is in imminent danger of serious bodily injury or death
- There is enough descriptive information about the victim and the abduction for law enforcement to issue an AMBER Alert to assist in the recovery of the child
- The abduction was of a child aged 17 years or younger.

The National Weather Service is responsible for issuing warnings when hazardous weather poses an immediate threat to life or property. However, the process of issuing a warning varies depending on the type of hazard. When weather hazards are small in size and develop quickly, forecasters must rely heavily on observational data from Doppler radar, satellite and other ground-based equipment and sensors. For hazards larger in scale and slower evolving the process hinges on forecasters' use and interpretation of numerical weather and statistical models. Once forecasters have high enough confidence there will be significant impacts, a warning is issued as far in advance as possible, for areas that are expected to see hazardous weather.

Mobile Device Alert Restrictions

Participating carriers are allowed to provide the capability for a user to opt out of the reception of a warning issued through the WEA with the exception of National alerts. Not all mobile devices are WEA-capable and participation in WEA by wireless carriers is widespread but voluntary. Some carriers may offer WEA over all or parts of their service areas. Even if a user has a WEA-enabled device, it will not receive a WEAs in a service area where the provider is not offering it.

A warning issued through the WEA takes precedence over any other application currently running on a mobile device. This means that, unless the participating carrier allows for the subscriber to block all alerts, once an alert is issued, it will be displayed and pause any other activity.

Existing Commercial Mobile Alert Applications

There are several applications in the market that relate the location of a mobile device to issue location specific notifications. Several methods such as geolocation or geofencing are used to achieve this. Geolocation relates the current location of the user to push specific notifications. In geofencing, the notification is pushed when the user is within the boundaries or "fences" of a particular location.

These notifications are subscription based, that is, a user is required to register (and in some cases, pay a subscription fee) and opt in the reception of location specific messages.

Here are some examples of mobile applications using Location-Based Push notifications:

Life360

The application provides notifications regarding the location of a user. It relates the location of the mobile device to a predetermined geographic location to provide alerts regarding the arrival of the user to the area.

AccuWeather

It provides specific weather alerts based on the mobile's location. The user can configure the type of alerts to receive.

FEMA: The Best Disaster App

It provides real-time weather and emergency alerts based on a pre-programmed location. It also provides directions for the closest emergency shelter.

GasBuddy

Provides gas prices of the nearby gas stations based on the location of the phone.

Conclusion

The issuing of an alert on a mobile device is restricted to widespread areas such as cities or regions and is only issued in cases of safety concerns covering these areas. These alerts are distributed by the Federal Communication Commission and are exclusively used for public safety situations. Cellphone carriers are allowed to opt out of the broadcast of these messages and provide the capability to the user to block them on their mobile devices.

The implementation of a capability for the RCVW system to provide an alert to mobile devices in the vicinity of a crossing when a train is approaching using existing emergency alert systems is not possible at this time. The service is restricted to national or regional emergencies.

A location-based application could be developed leveraging the Global Positioning Technology of the mobile device or by pre-programming a crossing location to provide alerts when the crossing is active. The RCWV Cloud Based Subsystem could provide the necessary crossing information (location and status) for a mobile device to produce an alert when the warning devices are activated. The effectiveness of this tool is dependent on market acceptance and widespread use.

References

- 1.- U.S. Department of Justice (2024). AMBER Alert. America's Missing: Broadcast Emergency Response. <https://amberalert.ojp.gov/>
- 2.- Federal Communications Commission (2024). Wireless Emergency Alerts (WEA). <https://www.fcc.gov/consumers/guides/wireless-emergency-alerts-wea>
- 3.- Life360 Inc (2024). <https://www.life360.com/>
- 4.-AccuWeather Inc (2024). <https://www.accuweather.com/>
- 5.- U.S Department of Homeland Security (2024). FEMA Mobile Products. <https://www.fema.gov/about/news-multimedia/mobile-products>
- 6.- GasBuddy LLC (2024). <https://www.gasbuddy.com/>